

# Complete Campus Dining & Pre-Order System Guide

A plain-English operational and architectural reference for students, canteen franchisees, university dining officials, and engineers.

## 1. What is V FOODS?

**V FOODS** is a high-speed campus pre-ordering, prepaid digital wallet, and kitchen management system built for collegiate food courts and large festival stalls. It eliminates physical counter lines by separating payment from ordering.

### Pre-Order Before Arriving

Students browse open campus canteens, select dishes, and order hot meals before walking over.

### Zero-Latency Digital Wallet

Checkout completes atomically in under 20ms from a pre-loaded balance, with 0% gateway dropouts.

### Kitchen Display System (KDS)

Kitchen cooks receive pre-orders instantly on a live digital Kanban board, preparing plates in advance.

### 4-Digit Token Pickup

Students show a 4-digit token or digital QR code at the counter, pick up hot food in seconds, and leave.

## 2. Full-Mode System Architecture

The architecture combines a fast, globally cached Progressive Web App (PWA) with a Python FastAPI backend and Supabase PostgreSQL with real-time WebSockets:

### SYSTEM COMMUNICATION PIPELINE

### 1. Student PWA

Menu browse, pre-order & instant token pass

### 2. Kitchen Staff KDS

Live Kanban ticket board & token scanner

### 3. Canteen Owner

Revenue ledger & menu pricing control

### 4. Super Admin

Campus-wide GMV & audit CSV export

## Secure HTTPS & Realtime WebSockets

### Render Global CDN

Vite + React 19 Frontend (Offline Service Worker)

### FastAPI Python Backend

PhonePe webhook verifier & token generator

## Row-Level Security & Atomic RPC

### Supabase PostgreSQL Cloud Database

PostgreSQL 15 · Row-Level Security (RLS) · Realtime Socket Bus · Atomic Stored Procedures

### PhonePe & MSG91

UPI Sandbox / Production & WhatsApp Alerts

### 3. The 4-Step Pre-Order Lifecycle

Every order follows an automated 4-stage pipeline that guarantees zero waiting line congestion:

| STAGE        | WHAT HAPPENS                                                         | STUDENT SCREEN                                       | KITCHEN KDS SCREEN                               |
|--------------|----------------------------------------------------------------------|------------------------------------------------------|--------------------------------------------------|
| 1. Placed    | Order placed from wallet. Database triggers instant WebSocket alert. | Shows 4-digit token (#4826) and estimated prep time. | Yellow ticket appears on KDS. Audio chime plays. |
| 2. Preparing | Kitchen staff taps "Start Cooking" and begins meal assembly.         | Live status bar advances to "Preparing".             | Blue ticket with live elapsed cooking timer.     |
| 3. Ready     | Staff packs meal and taps "Mark Ready". System sends push & SMS.     | Phone rings: "Order Ready for Pickup at Counter!"    | Emerald green badge in the pickup queue.         |
| 4. Collected | Student shows token at counter. Staff hands hot meal and marks done. | Order completed. Rating prompt appears.              | Order archived to daily sales report.            |

### 4. Wallet-First vs Traditional Payment

#### Why V FOODS Uses a Wallet-First Model

Traditional food ordering apps run credit cards or UPI during checkout. When 400 students try to pay simultaneously during break, bank servers fail. By topping up a campus wallet in advance, checkout completes in under 20 milliseconds with zero payment dropouts.

### 5. The Four Specialized Dashboards

| ROLE          | TARGET DEVICE                 | PRIMARY GOAL                       | KEY FEATURES                                                                               |
|---------------|-------------------------------|------------------------------------|--------------------------------------------------------------------------------------------|
| Student       | Smartphones                   | Pre-order food & skip queues       | Food Court Explorer, 3-column dish grid, wallet recharge, digital token collection pass.   |
| Shop Staff    | Kitchen tablet / touch screen | Cook & dispatch plates fast        | Live Kanban KDS, 1-tap status advance, instant "86" Sold-Out switch, token QR scanner.     |
| Canteen Owner | Laptop / Tablet               | Manage stall business & staff      | Daily gross & net sales, menu catalog & price editor, slot capacity caps, staff roster.    |
| Super Admin   | Desktop / Office              | Campus oversight & financial audit | Campus-wide GMV, 13 canteen health grid, Festival Mode switch, Financial Audit CSV export. |

## 6. Campus Canteens & Food Courts

VFOODS structures the physical dining counters across 5 collegiate pavilions at VIT Chennai:

|                                                                                                                                             |                                                                                                                                          |
|---------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Gazebo Food Court (4 Stalls)</b></p> <p>Gazebo C1 (Fast Food), Gazebo C2 (Desserts), Dakshin Chitra (Meals), Lassi House (Shakes)</p> | <p><b>North Square Food Court (4 Stalls)</b></p> <p>Georgia Canteen, Alpha Non-Veg Kitchen, Sri's Fast Food, Juice &amp; Rice Corner</p> |
| <p><b>AB3 Food Complex</b></p> <p>AB3 Amphitheatre Kitchen (Central hot snacks &amp; combo meals)</p>                                       | <p><b>Academic Blocks &amp; Pavilion</b></p> <p>AB1 Walkway, Delta Snacks, Aavin Milk Booth, and V Mart Store</p>                        |